

MODULE 12

Spine Trauma CT & MRI: Classification Systems & Instability

AO Spine · Denis Three-Column · TLICS · SLIC · Jefferson · Hangman · PLC Integrity

1.0 AMA PRA Cat 1

Spine

CME Article Review

LEARNING OBJECTIVES

1. Apply the AO Spine thoracolumbar injury classification (A, B, C types).
2. Use the Denis three-column model to assess fracture stability.
3. Calculate TLICS (thoracolumbar) and SLIC (subaxial cervical) scores for surgical decision-making.
4. Assess posterior ligamentous complex (PLC) integrity on MRI (STIR sequences).
5. Classify Jefferson (C1) and Hangman (C2) fractures using Levine-Edwards criteria.
6. Recognize clinical syndromes: anterior cord, central cord, Brown-Séquard on MRI.

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1. AO Spine Thoracolumbar Classification (A–C)

The AO Spine classification system is the current standard for thoracolumbar fracture classification, superseding the Denis system in research and many clinical centers. It evaluates morphology (Type A/B/C), neurological status (N0–N4), and case-specific modifiers (M).

AO Spine Thoracolumbar Fracture Types

Type	Category	Key Feature	Stability
A0	No or minor injury	Transverse process/spinous process; or contusion without bony injury	Stable
A1–A3	Compression injury	A1 = wedge; A2 = split/pincer; A3 = incomplete burst (anterior cortex only)	A1–A2 stable; A3 borderline
A4	Complete burst	Both endplates; middle column involvement; posterior wall retropulsion	Unstable if + PLC disruption
B1	Posterior ligamentous tension band disruption	PLC disrupted (MRI); transosseous = B2 (Chance)	Potentially unstable
B2 (Chance)	Posterior osseous tension band	Transosseous horizontal fracture; all 3 columns	Unstable — surgical usually
B3	Anterior disruption in hyperextension	Extension-distraction; anterior longitudinal ligament torn	Unstable
C	Translation/rotation	Displacement in any direction; complete instability	Highly unstable — surgical

2. Denis Three-Column Model & TLICS Scoring

Denis Three-Column Model

Column	Contents	Fracture Role
Anterior	Anterior 2/3 of vertebral body; anterior anulus; ALL	Compression fractures involve anterior column
Middle	Posterior 1/3 of vertebral body; posterior anulus; PLL	Critical column — disruption = burst; instability if + posterior
Posterior	Posterior bony arch (pedicles, facets, lamina); PLC (interspinous, supraspinous, lig. flavum, facet capsules)	Disruption = instability; PLC integrity assessed on MRI

TLICS Scoring — Thoracolumbar Injury

Component	Finding	Score
Injury Morphology	Compression fracture (wedge, burst)	1 (compression), 2 (burst)
	Distraction injury (ligamentous or osseous)	3
	Translation/rotation (subluxation, dislocation)	3
Posterior Ligamentous Complex (PLC)	Intact	0
	Indeterminate (equivocal on MRI/CT)	1
	Disrupted	2
Neurological Status	Intact	0
	Nerve root injury	2
	Cord injury (complete or incomplete)	2 or 3

KEY POINT

TLICS score: ≤ 3 = conservative; 4 = ambiguous (surgeon preference); ≥ 5 = surgical intervention recommended. The PLC component is the most important — disrupted PLC alone adds 2 points and substantially changes management.

3. Cervical Spine — Jefferson, Hangman & Cord Syndromes

Jefferson C1 Burst Fracture — Stability

Finding	Stable	Unstable
Transverse ligament status (MRI)	Intact; ADI normal	Disrupted; T2 high signal in ligament
Spence rule (bilateral overhang sum)	< 6.9 mm total on CT	≥ 6.9 mm = transverse ligament rupture likely
ADI (atlantodens interval)	≤ 3 mm (adult)	> 3 mm (adult); > 5 mm (child)

Levine-Edwards Hangman Fracture Classification

Type	Displacement	Angulation	DISc	Management
I	< 3 mm	$< 11^\circ$	Intact	Rigid collar; stable
II	> 3 mm	$> 11^\circ$	Disrupted	Halo or surgical
IIA	Minimal	Severe (atypical)	Disrupted	Halo (NO traction — risk of distraction)

Type	Displacement	Angulation	DISC	Management
III	Severe	Severe	Disrupted + facet Fx-dislocation	Surgical

Spinal Cord Injury Syndromes — MRI Correlation

Syndrome	MRI Finding	Deficit Pattern
Central cord	T2 hyperintensity in central cord (gray matter)	Hands > arms > legs weakness; bladder dysfunction; most common
Anterior cord	T2 signal in anterior cord	Motor lost + pain/temperature lost; proprioception PRESERVED
Brown-Séquard	T2 signal in hemicord (unilateral)	Ipsilateral motor + proprioception lost; contralateral pain/temp lost
Posterior cord	T2 signal in posterior cord	Proprioception/vibration lost; motor and pain/temp preserved
Complete SCI	Multi-level T2 signal; cord expansion or transection	Complete motor and sensory loss below injury level

CRITICAL

Type IIA Hangman fracture — do NOT apply traction (will further distract and injure). Use halo immobilization in slight compression. This is a critical management distinction from Type I/II which tolerate traction.

References

- Vaccaro AR et al. AO Spine thoracolumbar classification. Spine 2013.
- Denis F. Spine 1983.
- Levine AM, Edwards CC. Traumatic spondylolisthesis of C2. J Bone Joint Surg Am 1985.